Sample Exam

Medical Image Analysis

Student name:____

Immatriculation number:____

Exam Questions

Note: All the questions have equal weight.

1. In 2D acquisitions, slices are acquired in one pre-defined plane and stacked along the remaining direction: axial slices stack head-to-toe, sagittal slices left-to-right, coronal slices anterior-to-posterior. Voxel sizes are given as in-plane (two values) and through-plane (the slice thickness), and the field-of-view is image size $\times$ voxel size.

   You are handed a volume with the following properties, but the acquisition plane is not recorded: field-of-view 240 mm head-to-toe, 240 mm left-to-right, 180 mm anterior-to-posterior; voxel size in-plane 0.75×0.75 mm^2, through-plane 4.5 mm.

   (a) In which plane was this volume acquired? Justify your answer from the numbers.

   (b) How many voxels does the volume have in the head-to-toe, left-to-right and anterior-to-posterior directions, respectively?

   (c) The volume is resampled to isotropic $1.5 \times 1.5 \times 1.5$ mm^3 voxels. Give the resize factors and the resulting image size, and verify that the field-of-view is unchanged.

2. In bilinear interpolation the intensity at a query point x_1 (with $x_1^0 \le x_1 \le x_1^1$) is first interpolated along one dimension from the two neighbouring pixels. Which of the two expressions below is the correct linear interpolation? Please circle the correct choice.

$$I(x_1) = \frac{x_1^1 - x_1}{x_1^1 - x_1^0}\, I(x_1^0) + \frac{x_1 - x_1^0}{x_1^1 - x_1^0}\, I(x_1^1) \qquad\qquad I(x_1) = \frac{x_1 - x_1^0}{x_1^1 - x_1^0}\, I(x_1^0) + \frac{x_1^1 - x_1}{x_1^1 - x_1^0}\, I(x_1^1)$$

3. Probabilistic reasoning underlies most of this course.

   (a) If each pixel intensity is treated as a realization of a random variable, the (normalized) histogram and the cumulative histogram of an image approximate two standard probabilistic objects. Which two — and what assumption about the pixels makes this approximation valid?

   (b) The Maximum-A-Posteriori estimate is $\arg\max_l\, p(i \mid l)\, p(l)$ and the Maximum Likelihood estimate is $\arg\max_l\, p(i \mid l)$. Under what condition on the prior are the two identical?

4. Non-local means computes $\hat{I}(x) = \sum_y w(x,y)J(y)$ with weights from the similarity of the patches $N(x)$ and $N(y)$, a degree-of-filtering parameter h, and averaging restricted to a search window $W(x)$. Please indicate whether the statement is true or false by circling the correct choice in each of the following.

   (a) True False Choosing the patch to be the entire image, $N(x) = \Omega$, reduces non-local means to Gaussian filtering.

   (b) True False Choosing the patch to be the single pixel, $N(x) = \{x\}$, reduces non-local means to Yaroslavsky's filter, a special case of bilateral filtering.

   (c) True False Increasing h decreases the amount of smoothing.

   (d) True False The averaging is restricted to a search window $W(x)$ rather than the whole image mainly to reduce the computational cost.

5. A 2D landmark-based registration uses thin-plate splines with the $N = 50$ landmarks as control points, combined with a global affine part. How many parameters does the transformation have? How does this compare with a naive deformable parameterization of a 256×256 image that stores one independent displacement vector per pixel?

6. An Active Appearance Model extends a statistical shape model with a model of texture. Please describe briefly (i) how the texture model is built from the training images, and (ii) how shape and texture are combined into a single set of appearance parameters. You do not need to write down the equations.

7. Which of the following statements about atlas-based segmentation practice is **inaccurate**?

   ◯ (A) An atlas consists of an image together with its manual segmentation; it is registered to the unseen patient image and the labels are propagated along the transformation.

   ◯ (B) When an average (mean) atlas is built from a population, the choice of the reference subject does not influence the result — averages built from different references are the same.

   ◯ (C) The quality of an atlas-based segmentation depends on the image similarity between atlas and patient, the presence of pathology, the quality of the registration, and how typical the patient is of the atlas population.

   ◯ (D) Shape-based averaging fuses several segmentations by averaging their distance transforms rather than voting on their labels.

8. Which of the following statements on uncertainty estimation is **incorrect**?

   ◯ (A) One source of uncertainty is that the input x may not uniquely identify the label y.

   ◯ (B) Another source is that the training set D is itself a random sample from the population, and a different draw would have produced different parameters.

   ◯ (C) The Generalized Energy Distance can be evaluated on an ordinary test set with a single annotation per image, which is why it is the most general evaluation of uncertainty.

- ◯ (D) The Expected Calibration Error measures the average gap between the confidence a model assigns and the accuracy it actually achieves at that confidence.

9. Which of the following statements from the machine-learning recap is **incorrect**?
   - ◯ (A) Hyper-parameters — the architecture, the stopping point of the optimization, the learning rate — are chosen using a validation set, ideally as a nested optimization inside the training of the parameters.
   - ◯ (B) The training loss is a sum over samples, so the optimal parameters do not have to satisfy any single sample perfectly — they satisfy the samples on average.
   - ◯ (C) The prediction error on the test set must be computed with the same differentiable loss that was used for training, since otherwise the numbers are not comparable.
   - ◯ (D) For a new sample the prediction is $y \approx f(x;\theta^*)$, and in general $y \neq f(x;\theta^*)$.

10. "Meaningful perturbation" explains a classifier's decision by perturbing the input. Which of the following statements describes the method correctly?
    - ◯ (A) It searches the training set for the image that most strongly activates the class and presents it as the explanation.
    - ◯ (B) It optimizes an information-deletion mask — applying blur, noise or masking — that covers as little of the image as possible while dropping the class score as much as possible.
    - ◯ (C) It requires no evaluations of the model, which makes it the cheapest explanation method.
    - ◯ (D) It computes the gradient of the class score with respect to the convolutional feature maps and pools it into a heatmap.
    - ◯ (E) Because the mask is optimized, the resulting perturbations are guaranteed to remain natural-looking.

11. A classification network takes grey-scale input images of size 68×68 with one input channel. All convolutions are *valid* (no padding) with stride $S = 1$:
    - **Conv1**: 6 kernels with $K = 5 \times 5$ **Pool1**: max pooling 2×2, $S = 2$
    - **Conv2**: 12 kernels with $K = 3 \times 3$ **Pool2**: max pooling 2×2, $S = 2$
    - **Conv3**: 24 kernels with $K = 3 \times 3$

    Circle True or False:

    (a) True False The output of Conv3 has a spatial size of 15×15.

    (b) True False The global receptive field of a neuron in Conv3 is 20×20.

    (c) True False The global stride of a neuron in Conv3 is 8×8.

    (d) True False The network's convolutional layers have $5 \cdot 5 \cdot 1 \cdot 6 + 3 \cdot 3 \cdot 6 \cdot 12 + 3 \cdot 3 \cdot 12 \cdot 24 = 3\,390$ kernel weights, and including one bias per kernel the total number of parameters is $3\,432$.

12. Which of the following statements are correct (multiple answers are possible)?
    - ◯ (A) In FCN-style segmentation, outputs taken at several depths of the network are upsampled by different factors and combined, so that fine detail from shallow layers complements the context of deep layers.
    - ◯ (B) A label map may be transformed with bilinear interpolation provided the interpolated values are rounded to the nearest integer label afterwards.

○ (C) Matching the mean and standard deviation of two images, $f(j) = (j - \mu_j)\,\sigma_i/\sigma_j + \mu_i$, matches second-order statistics and is less sensitive to outliers than min–max normalization.

○ (D) A similarity transformation has 4 degrees of freedom in 2D and 7 in 3D.

○ (E) The curl of a transformation field, $\nabla \times T$, quantifies the local volumetric change induced by the transformation.